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MINISTRY OF ROADS AND TRANSPORT

RDM 6.4

Road Design Manual

Volume 6: Traffic Control Facilities & Communication Systems Design

Part 4: Other Traffic Control Devices (*Variable Message Signs*)

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Foreword

This manual was developed by the Ministry pursuant to The Fourth Schedule of the Constitution which assigns to the National Government the functions and powers of setting standards for the construction and maintenance of all public roads including those under the County Governments.

It is part of a series of manuals that replace the first generation of road manuals developed in the first and second decades after independence. This second generation of the road manuals were developed to cover the entire road project cycle covering planning, appraisal, design, contracts, construction, maintenance, operations and monitoring. The series incorporates best practices, climate change considerations, and recent technologies to enable the provision of road infrastructure that is safe, secure, and efficient.

Under the Kenya Vision 2030 long term plan, infrastructure expansion and modernisation are some of the foundations for the realisation of economic, social and political transformation of Kenya into a rapidly industrialising middle-income country. The plan envisages an integrated, safe and efficient transport and communication infrastructure network consisting of roads, railways, ports, airports, waterways, and telecommunications infrastructure.

The strategies to be pursued under the Vision 2030 plan to improve infrastructure services and to maximise the economic and social impacts of infrastructure development and management include: Strengthening of the institutional framework for infrastructure development and maintenance; Raising efficiency and quality of infrastructure projects; Enhancing local content of identified infrastructure projects to minimise import content; Benchmarking infrastructure facilities and services provision with globally acceptable performance standards; and, Implementing infrastructure projects that will stimulate demand in hitherto marginalised areas.

The first three 5-year Medium Term Plans (MTP) under the Vision 2030 from 2008 to 2022 targeted construction of 1,950 km, 5,500 km and 10,000 km of new paved roads under MTP I, II and III, respectively, totalling 17,450 km. This was a massive infrastructure development program intended to double the paved road network in 10 years compared to 8,600 km developed from independence in 1963 to 2008.

Implementation of MTP I to III resulted in the construction of 14,000 km of paved roads, which extended the paved road coverage to Arid and Semi-Arid regions, that had been previously neglected. However, some key milestones of the Vision 2030 goals have not been realised. This has been due to internal and external challenges. External challenges included: climate change – prolonged droughts; the emergence of COVID-19 pandemic; global supply chain disruptions; exchange rate volatility; and rising interest rates in the leading economies.

The internal challenges included: inadequate road maintenance equipment; pavement overloading by heavy goods vehicles; huge maintenance backlog of the road network; low contracting and supervision capacity particularly in the Counties; poor quality control and assurance of works; congestion in urban areas; encroachment on road reserves; high costs and delays in payments of land acquisition; lack of harmonisation of cross-border transport regulation and operational procedures; rapid urbanisation; increased traffic volume with exponential growth of motorcycle traffic; high cost/delays in relocation of utilities and services along and across road reserves; inadequate funding of projects and programs; and, delay or default in payments for goods, services and works.

The implementation of MTP III came to an end on 30th June 2023, ushering in the implementation of the Fourth Medium Term Plan (MTP IV), which has been aligned to the aspirations of the Kenya Vision 2030 and the Kenya Kwanza Government's Bottom-Up Economic Transformation Agenda (BETA) planning approach and its key priorities.

BETA is the Government's transformation agenda geared towards economic turnaround through a value chain approach. BETA has targeted sectors with the highest impact to drive economic recovery and growth. This will be achieved through bringing down the cost of living; eradicating hunger; creating jobs; expanding the tax base; improving foreign exchange balances; and inclusive growth. BETA ensures rational resource allocation by eliminating wastage of resources occasioned by duplication, overlaps, fragmentation and ineffective coordination in the implementation of programmes and projects.

The Fourth Medium Term Plan key priorities are clustered under five key sectors, namely: Finance and Production; Infrastructure; Social; Environment and Natural Resources; and Governance and Public Administration. The infrastructure sector seeks to: enhance transport connectivity by constructing 6,000 km of new roads, maintaining rural and urban roads, rail, air and seaport facilities and services; expand communication and broadcasting systems; and promote the development of energy generation and distribution by increasing investments in green energy (geothermal, wind, solar and hydro). The infrastructure gap is expected to be bridged by promoting economic participation of the private sector through public private partnerships in the financing, construction, development, operation, and maintenance of infrastructure.

The plan entails a shift of focus to fundamentals in project planning and implementation which include: respect for technical input, regulations and standard practices; adherence to project life cycle i.e., planning, feasibility studies and design before procurement of works; public and stakeholder consultation; procurement within budgetary ceilings; shifting focus during project implementation from the finished product 'black top' to the construction of the foundation; building local capacity particularly MSMEs by ensuring prompt payments; and capacity building at all levels to enable internalisation of policies and processes.

The first generation of the road manuals were used for 35 to 45 years. It is my sincere hope that the second generation of the road standards which have been developed in alignment with the Government's strategy will provide guidance in solving most of the above challenges and those expected to emerge in the next 50 years. Implementation of the manuals will enable achievement of the Government aspirations which include inclusive growth; creation of sustainable employment; building of MSMEs; climate change adaptation and realisation of the UN SDGs; enhanced efficiency in management of infrastructure and transport system; and, laying the foundation for the next national long-term plan at the end of the Vision 2030.

On behalf of the Government of Kenya, I would wish to thank the European Union for financing the development of the first drafts of the manuals in 2009 and the African Development Bank for the financial support in the review and updating of the manuals. I would also like to thank the members of the National Steering Committee and the Technical Task Force for their input. The Technical Administrators, and the Kenya National Highways Authority (KeNHA) for the procurement and able administration of the consultancy Contract. I also thank the Consultant, TRL Limited for their role in providing technical expertise that was essential for the success of the manuals updating exercise. I also wish to express my deepest appreciation to our stakeholders and all those who have contributed to this process and the staff of the Ministry for their continued input.

Hon. Davis K. Chirchir, E.G.H

Cabinet Secretary, Ministry of Roads and Transport

Preface

The Manual focusses on traffic control devices for speed reduction measures on roads and highways. It details out speed breakers, circular humps, trapezoidal humps, speed bumps, speed tables / raised pedestrian crossing and thermoplastic bar marking.

The Manual details the variable message signs (VMS) that can be used to control traffic. Unlike the static traffic control devices like the road signs and pavement marking, the contents of the VMS can be changed from situation to situation. VMS can be placed at important locations in the form of an overhead sign. Portable VMS can be moved to different locations as needed.

The Manual explains the principles of VMS. The main purpose of VMS is to control, warn and inform traffic about incidents, diversions, road works, weather and special conditions. The content, timing, letter size and length are explained.

For static VMS, the location, display and language to be used are explained. Portable VMS are to be placed at suitable locations and correct information display is important.

Mechanical components, mounting structures, equipment, controller, electronics and communication are explained. Design life of VMS is 10 years. It generally needs a central computer. The requirements of power and storage are explained. Design process of the VMS display and sample messages are provided.

Eng. Joseph M. Mbugua, CBS

Principal Secretary, Ministry of Roads and Transport

Document Management

Document Status:

This document has the status of a Manual. Users shall apply the contents there-in to fully satisfy the requirements set out. The content of the manual is based on current practice in Kenya and latest practices in the road sector, both regionally and internationally.

Sources of the Document:

Copies of the document can be obtained from:

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Abbreviations

AASHTO	American Association of State Highways and Transportation Officials
AM	Ante Meridiem
ATMS	Advanced Traffic Management System
Caltrans	California Department of Transportation
CDMA	Code Division Multiple Access
CPU	Central Processing Unit
DC	Direct Current
DfT	Department for Transport, United Kingdom
EMI	Electromagnetic interference
FHWA	Federal Highway Administration
FT	Fixed Time
GSM	Global System for Mobile Communication
Hz	Hertz
I/O	Input / Output
IP55	Ingress Protection 55
ITS	Intelligent Transportation System
Kmph	Kilometres per hour
km/hr	Kilometres per hour
LDR	Light Dependent Resistor
LED	Light Emitting Diode
MB	Mega Byte
MHz	Mega Hertz (Million Hertz)
MIS	Management Information System
NMT	Non-motorised traffic
NTCIP	National Transportation Communication for ITS Protocol
PBU	Push Button Unit
PCU	Passenger Car Unit
PM	Post Meridiem
PMPP	Point to Multi-point communication Protocol
PPH	Passengers per Hour
PSTN	Public Switched Telephone Network
QH	Quartz Halogen
S	Stroke
SRAM	Static Random Access Memory
TFT	Thin Film Transistor
USDOT	United States Department of Transportation
USB	Universal Serial Bus
V	Volts
VA	Vehicle Actuated
VMS	Variable Message Sign
VPH	Vehicles per Hour
W	Watt

Glossary of Terms

Acceleration Lane	A paved auxiliary lane, including tapered areas, allowing vehicles to accelerate when entering the through-traffic lane of the roadway.
Advance Warning Signals	An active warning device consisting of a warning sign with Alternating Flashing Amber displays to warn approaching motorists of their imminent arrival at a traffic signal installation.
Algorithm	A set of commands that must be followed for a computer to perform calculations
Approach	That section of road, consisting of one or more lanes, used by vehicles approaching an intersection or mid-block site.
Capacity	The maximum sustainable flow rate at which vehicles or persons reasonably can be expected to traverse a point or uniform segment of a lane or roadway during a specified time period under given roadway, geometric, traffic, environmental, and control conditions; usually expressed as vehicles per hour, passenger cars per hour, or persons per hour.
Expressway	A multilane, divided highway with a minimum of two lanes for the exclusive use of traffic in each direction and control of access without traffic interruption.
Queue	A line of vehicles or pedestrians waiting to proceed through an intersection. Slowly moving vehicles or pedestrians joining the back of the queue are usually considered part of the queue. The internal queue dynamics can involve starts and stops. A faster-moving line of vehicles is often referred to as a moving queue or a platoon.
Road Marking	A word, figure, symbol, mark, line, raised marker or stud, or something else, on the surface of a road to direct or warn traffic, but does not include a Painted Island.
Sight Distance	The distance over which a road user needs to have unobstructed sight to respond to a visual cue, or to safely avoid a conflict.
Signalised Crossing	An area of the road used by pedestrians when crossing the road with the guidance of pedestrian signals at a mid-block or intersection location and can be used by cyclists if bicycle signals are provided.
Speed	Distance travelled per unit time. In a time – distance diagram, the slope of the trace of a vehicle is its speed. Approach Speed is the uninterrupted (mid-block) cruise speed of vehicles before being affected by traffic signals. This can be represented by the speed limit. Negotiation Speed is the safe speed of a vehicle moving through the controlled area of the intersection. For turning vehicles, this can be determined as a function of the negotiation radius. Eighty-fifth Percentile Speed is the speed at or below which 85 per cent of the traffic travels.
Stop Line	A single continuous line marked across all or part of a road, behind which vehicles should stand when required to stop by traffic light signals or regulatory signs. Also see Give-Way Line.
T-Intersection	An intersection where two roads meet (whether or not at right angles) and one of the roads ends.
Traffic	Movement of vehicles and pedestrians (people and goods) along a route.
Traffic Composition	The proportions (usually expressed as a percentage) of the different vehicle classes and turning vehicles within the total traffic flow.

Glossary of Terms *(continued)*

Traffic Control Device	Any traffic sign, road marking, traffic signal, or other device, placed or erected under public authority for the purpose of regulating, directing, warning or guiding traffic.
Traffic Sign	A board, plate, screen, or another device, whether or not illuminated, displaying words, figures, symbols or anything else to regulate, direct or warn road users, and includes children crossing flag, a hand-held stop sign, a parking control sign and a variable illuminated message sign, but does not include traffic signals.
Traffic Signal	A signal that controls vehicle and pedestrian traffic at an intersection or on a road by means of Red, Amber, Green or white light displays, and includes circular and arrow signals, pedestrian signals, bicycle crossing signals, B (bus) and T (tram) signals, overhead lane control signals, and twin Red or Amber signals.
Traffic Volume	The number of vehicles or pedestrians passing a given point on a lane or carriageway during a specified period of time.
Upstream	In the direction opposite to the movement of traffic.
Walkway	That portion of a road reserved exclusively for pedestrians. Often referred to as footpath.

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1 Introduction

1.1 General

This Manual was prepared by the Ministry of Roads and Transport as part of a series of manuals that cover the entire project cycle. The series incorporate best practices, climate change considerations, and recent technologies thereby enabling the provision of road infrastructure that is safe, secure and resilient.

The Kenya road manual series is as follows:

Project Cycle Stage	Manual: Volume or Part/Chapter	Code
A. General	Procedures and Standards Manual	PSM
	1. General	
	2. Policies	
	3. Procedures Guidance	
	4. Codes of Practice	
	5. Guidelines	
B. Planning	Network and Project Planning Manual	NPM
	1. Road Classification	
	2. Route/Corridor Planning	
	3. Route/Corridor Planning	
	4. Highway Capacity	
	5. Project Planning	
C. Appraisal	Project Appraisal Manual	PAM
	1. Environmental Impact Assessment and Audit	
	2. Social Impact Assessment	
	3. Traffic Impact Assessment	
	4. Road Safety Audits	
	5. Project Appraisal	
D. Design	Road Design Manual	RDM
	1. Geometric Design	
	2. Hydrology and Drainage Design	
	3. Materials and Pavement Design for New Roads	
	4. Bridges and Retaining Structures Design	
	5. Pavement Maintenance, Rehabilitation and Overlay Design	
	6. Traffic Control Facilities and Communication Systems Design	
E. Contracts	Works and Services Contracts Manual	WSCM
	1. Forms of Contracts	
	2. Standard Specification for Road and Bridge Construction	
	3. Bills of Quantities	
F. Construction	Road Construction Manual	RCM
	1. Construction Management	
	2. Project Management	
	3. Site Supervision	
	4. Quality Assurance	
	5. Quality Control	

This table continues onto the next page...

Project Cycle Stage	Manual: Volume or Part/Chapter	Code
G. Maintenance	Road Asset Management Manual	RAAM
	1. Maintenance Management	
	2. General Maintenance	
	3. Pavement Maintenance	
	4. Bridges and Structures Maintenance	
H. Operations	Road Operation Manual	ROM
	1. Traffic Management	
	2. Vehicle Load Control	
	3. Emergency Services	
	4. Tolling	
I. Monitoring & Evaluation	Road Design Manual	MEM
	1. Performance Monitoring Manual	
	2. Technical Audits	
	3. Poverty, Gender Equality and Social Inclusion Monitoring	

This manual, as indicated below, is Volume 6, Part 4: Other Traffic Control Devices and is part of the Roads Design Manual (RDM) whose volumes, parts and coding system are shown in Table 1.1.

Table 1.1 Road Design Manual (RDM) Coding Structure

Vol.	Manual Title	Part Name	Code
1	Road Design Manual: Vol. 1 Geometric Design	Part 1 - Topographic Survey	RDM 1.1
		Part 2 – Traffic Surveys	RDM 1.2
		Part 3 – Geometric Design of Highways, Rural and Urban Roads	RDM 1.3
2	Road Design Manual: Vol. 2 Hydrology & Drainage Design	Part 1 – Hydrological Surveys	RDM 2.1
		Part 2 – Drainage Design	RDM 2.2
3	Road Design Manual: Vol. 3 Materials & Pavement Design for New Roads	Part 1 – Ground Investigations and Material Prospecting	RDM 3.1
		Part 2 – Materials Field and Laboratory Testing	RDM 3.2
		Part 3 – Pavement Foundation and Materials Design	RDM 3.3
		Part 4 – Flexible Pavement Design	RDM 3.4
		Part 5 – Rigid Pavement Design	RDM 3.5
4	Road Design Manual: Vol. 4 Bridges & Retaining Structures Design	Part 1 – Geotechnical Investigation and Design	RDM 4.1
		Part 2 – Bridge and Culvert Design	RDM 4.2
		Part 3 – Retaining Structures Design	RDM 4.3
		Part 4 – Reinforced Fill Structures Design	RDM 4.4
		Part 5 – Bridges and Structures Condition Survey	RDM 4.5
		Part 6 – Bridge Maintenance Design	RDM 4.6
5	Road Design Manual: Vol. 5 Pavement Maintenance, Rehabilitation & Overlay Design	Part 1 – Pavement Condition Survey	RDM 5.1
		Part 2 – Pavement Maintenance, Rehabilitation and Overlay Design	RDM 5.2
6	Road Design Manual: Vol. 6 Traffic Control Facilities & Communication Systems Design	Part 1 – Road Marking	RDM 6.1
		Part 2 – Traffic Signs	RDM 6.2
		Part 3 – Traffic Signals and Communication System	RDM 6.3
		Part 4 – Other Traffic Control Devices	RDM 6.4
7	Road Design Manual: Vol. 7 Road Lighting Design	Part 1 – Grid-connected Road Lighting	RDM 7.1
		Part 2 – Solar Road Lighting	RDM 7.2

This manual must be applied sensibly and flexibly in conjunction with the skill and judgement of the designer. Compliance with the guidance given in the manual does not relieve designers of the responsibility for establishing that their design is suitable, appropriate, safe, and adequate for the purpose stated in the project requirements.

1.2 Objective

The objective of the Manual is to guide users in design and usage of other traffic control devices. These include devices for speed reduction measures and variable message signs (VMS). Speed reduction measures are to be used for traffic calming. VMS are to be used for dynamic display of messages as per demand. In the portable version of VMS, the location of signs can be changed.

1.3 Scope

The Manual covers use of other traffic control devices like speed breakers, humps, bumps, speed tables and raised pedestrian crossings. Bar markings are a series of markings to alert the driver. The placing based on approach speeds is explained.

The Manual also covers Variable Message Signs. The purposes of traffic control, warning and information dissemination are explained. Usage of VMS for various types of incidents are detailed. Purpose, length, height, and type of messages to be used are provided.

The requirements for portable VMS are different from static VMS. Placing and messages of portable VMS are detailed.

Components including sign face, housing, access, controller cabinet, interfaces, power, requirement of central computer, storage and design of VMS are explained.

1.4 Organisation of the Manual

The Manual is structured in the following manner:

Chapter 1 gives the scope, objective, organisation of the manual and design flow.

Chapter 2 deals with speed reduction methods to warn drivers.

Chapter 3 deals with Variable Message Signs. It introduces the concept of VMS, why and when it is needed. Then it explains where and how it should be used, and what the contents of messages should be. It explains the various components of VMS and the design process for VMS deployment.

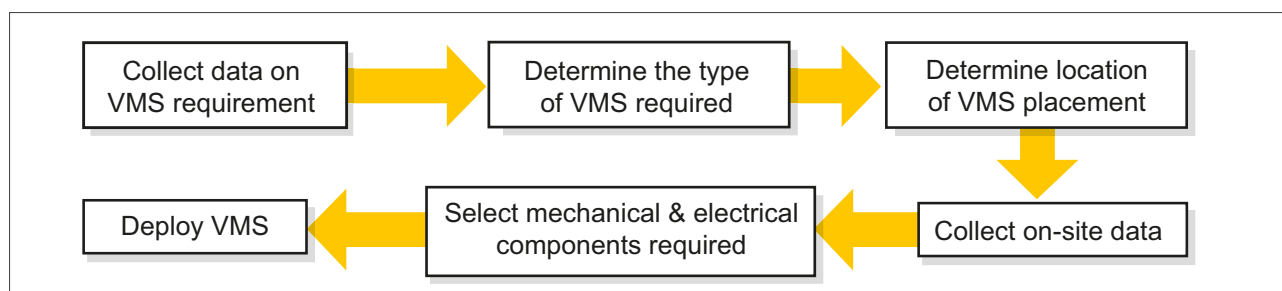
1.5 Design Process

The design process of using speed control measures is:

1. Analyse where speed reduction is necessary.
2. Based on the approach speed, reduction in speed required and the class of roadway, choose the type of hump required.
3. Include standard drawings in plan drawing for installation.
4. Prepare Bill of Quantities for installation.

The process of deploying VMS is shown in **Figure 1.1**.

Figure 1.1 VMS Deployment Process



2 Speed Reduction Measures

2.1.1 Speed Breakers

Speed breakers are used to warn drivers about hazards ahead in advance. Speed breakers are implemented in the form of speed humps or speed bumps. A speed hump is a raised area in the roadway pavement surface extending transversely across the carriageway. Speed humps are sometimes referred to as 'pavement undulations' or 'sleeping policemen'. Most agencies implement speed humps with a height of 76 to 90 mm and a length of 3.7 to 4.3 m.

A speed bump is also a raised pavement area across a carriageway. Speed bumps generally have a height of 76 to 152 mm with a length of 0.3 to 1 m.

From an operational standpoint, speed humps and bumps have critically different impacts on vehicles. Properly placed speed humps can reduce vehicle speed to about 30 kmph. A speed bump can cause driver discomfort. It typically reduces the vehicle speed to around 8 kmph or less at each bump.

Speed bumps of varying design are available for use on roadways and parking lots. Speed humps, on the other hand, have evolved from extensive research and testing and have been designed to achieve a specific result on vehicle operations without imposing unreasonable or unacceptable safety risks.

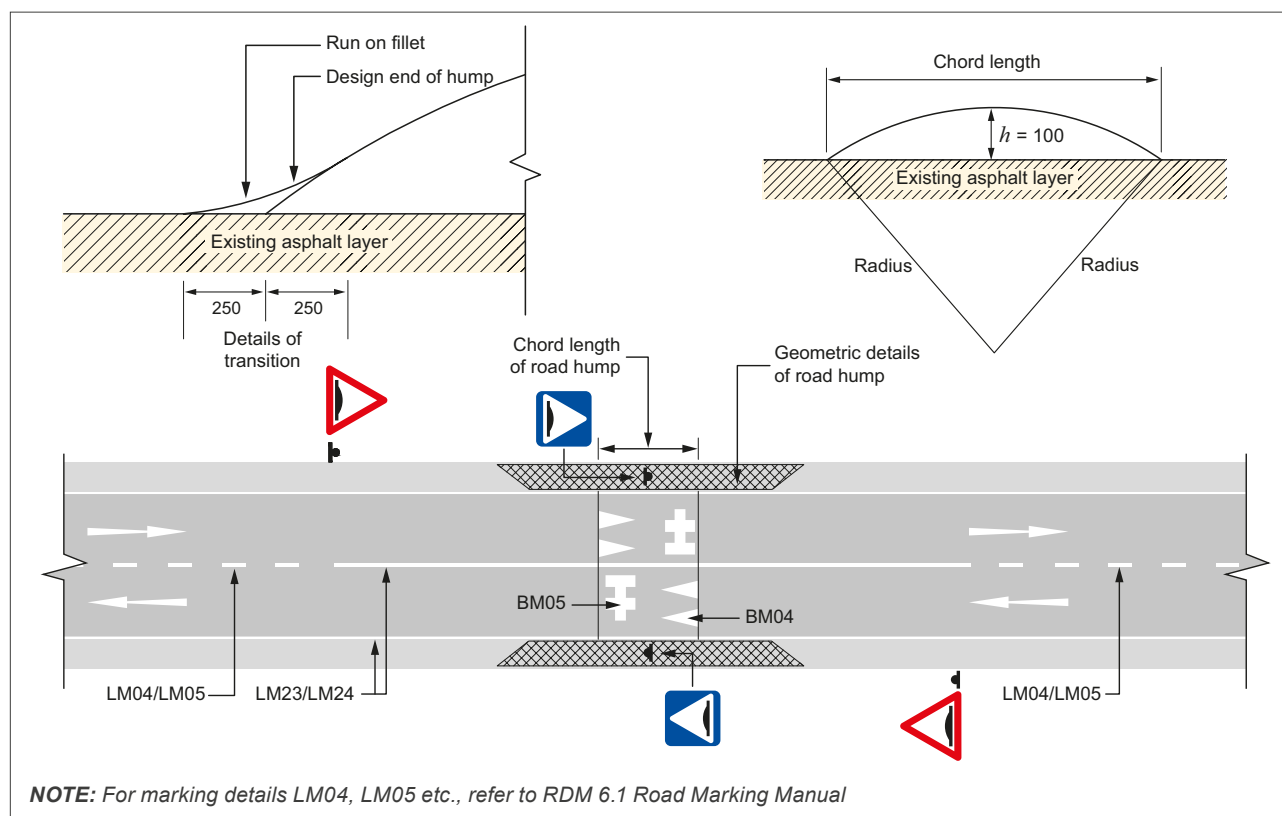
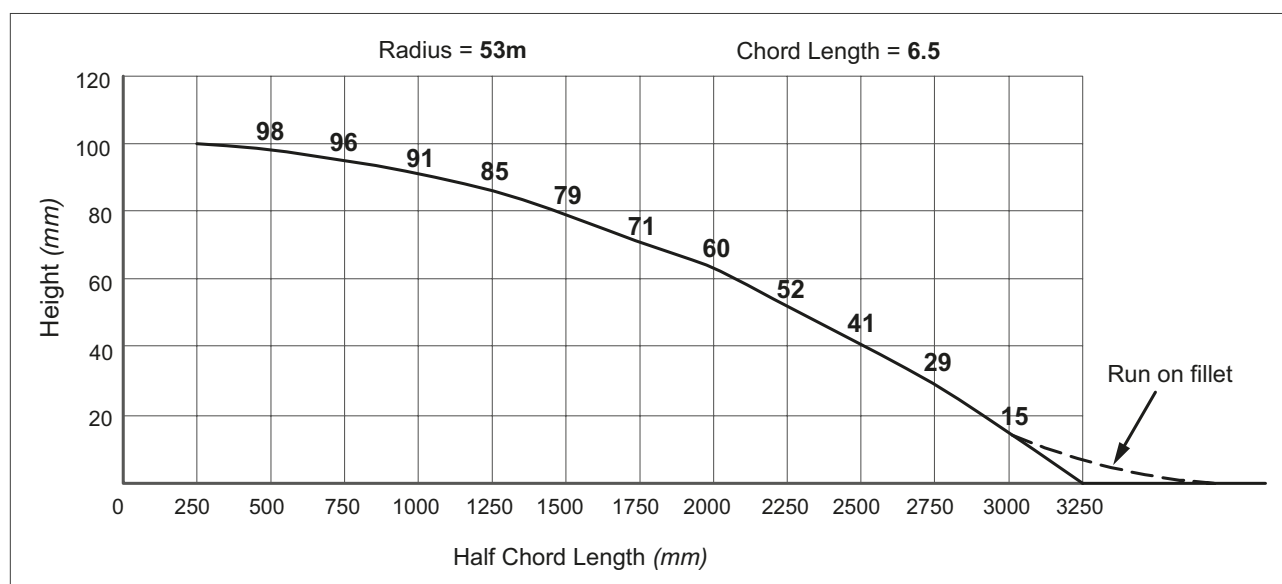
For better night time visibility of bumps and humps, they are marked with retro-reflective material and reinforced with road studs. Speed breaker markings are supplemented with warning signs in advance of pedestrian crossings and also informatory signs at the location of these crossings.

Figure 2.1 shows the types of marking to be used on the speed breakers. It comprises of two rows of chequered markings consisting of alternate black and white bands of 500 mm width on either side of tapering section. The triangular markings are applied in the case of both round top and flat top humps. The width of the base of the triangular marking is 750 mm and the height from its base to apex is a maximum of 1.85 m.

It is desirable to provide a raised walkway for at least a small length as shown in Figure 2.1. This measure would help in preventing traffic from using the shoulder in an attempt to avoid the road hump installed in the main carriageway. Road humps of various chord lengths can be developed, thereby facilitating different speed; the geometric details for construction of road humps have been illustrated in Figure 2.2.

2.2 Circular Humps

The profile of a circular hump is an arc with a radius varying from 11 m to 113 m and a chord length varying from 3 to 9.5 m. The desired rise of circular hump is 100 mm. Humps with a rise less than 100 mm will not be effective in speed reduction. Rises higher than 100 mm may cause damage to vehicles. For roads with bus traffic, Table 2.1 indicates the speeds at which buses can reasonably pass the individual humps. For buses to pass at 40 km/h, the radius of the hump must be 180 m and chord length 12 m. Figure 2.1, Figure 2.2 and Table 2.1 give the geometrical details of circular road humps.

Figure 2.1 Recommended Radii and Chord Lengths for Circular Humps (rise 100 mm)**Figure 2.2** Geometric Details of Circular Road Humps**Table 2.1** Geometric Details of Circular Road Humps

Desired Speed (km/h)	Radius (m)	Chord Length (m)	Bus Speed During Passage (km/h)
20	11	3.0	5
25	15	3.5	10
30	20	4.0	15
35	31	5.0	20
40	53	6.5	25
45	80	8.0	30
50	113	9.5	35

Source: Indian Roads Congress IRC 35: 2015

2.3 Trapezoidal Humps

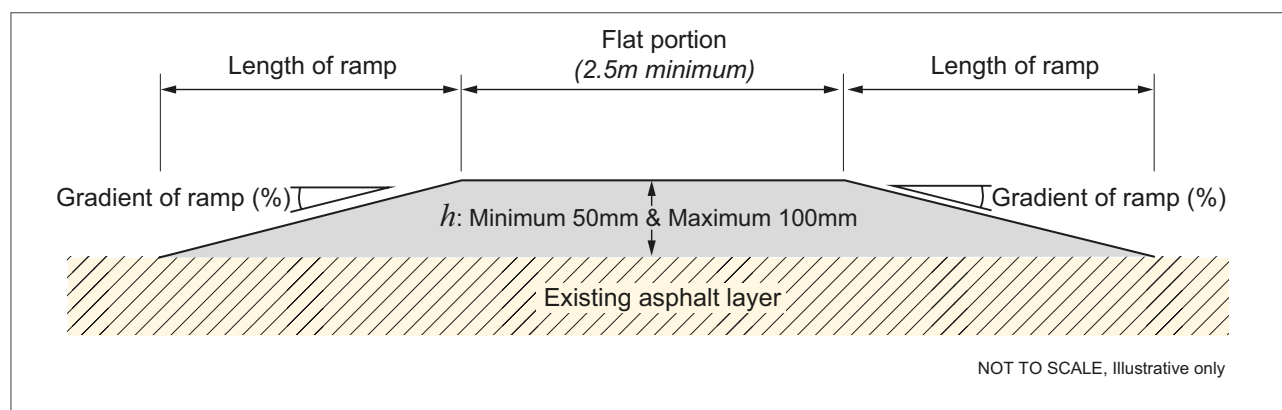
A trapezoidal hump consists of a 50 to 100 mm raised flat section with ramps on both sides. Cars driving over it have a moderate level of discomfort. Lorries and buses need to drive very slowly. Figure 2.3 and Table 2.2 give geometric details of trapezoidal humps for various speeds. This includes the speeds at which buses may pass without much discomfort. If it is intended for buses to pass at speeds of 35 and 40 km/h, ramp lengths must be 3.3 and 4.0 m, with gradients of 3.0 and 2.5 % respectively.

Table 2.2 Recommended Ramp Lengths and Gradients for Trapezoidal Humps

Desired Speed (km/h)	Length of Ramp (m)	Gradient (m)	Bus Speed During Passage (km/h)
20	0.7	14.0	Near zero
25	0.8	12.5	5
30	1.0	10.0	10
35	1.3	7.5	15
40	1.7	6.0	20
45	2.0	5.0	25
50	2.5	4.0	30

Source: Indian Roads Congress IRC 35: 2015

Figure 2.3 Geometric Details of Trapezoidal Road Humps



2.4 Speed Bumps

Speed bumps are similar to humps. Bumps have steeper rise and tighter curvature to increase their effectiveness to reduce speed. Bumps are made of plastic or rubber. These are manufactured in modular design so that they can be used for varying widths. Bumps are secured by passing a bolt through a pre-drilled hole in the plastic body. Bumps are used to bring down the vehicle speed to around 5 – 10 kmph.

2.4.1 Temporary Speed Bumps

Speed bumps can be of a temporary or permanent type. Temporary bumps can be moved to different locations. It can be a short-term fix before a more permanent solution can be implemented. Temporary bumps should be used as a part of work zone traffic management plan, with proper warning signs. The bumps should be clearly visible during day time and illuminated at night time. A temporary bump cannot be used if it poses a safety hazard.

2.4.2 Permanent Speed Bumps

Precast, ready to install permanent type road bumps are available. These bumps should not be installed on high-speed roads. The bumps get damaged by repeated strikes of traffic. The bolt remains on the pavement and can damage the tyres of vehicles. Therefore, its usage is only recommended for local and collector streets.

The requirements for installing permanent speed bumps are¹:

- Bumps should be installed on roads where the approach speed is less than 50 kmph.
- If the approach speed is more than 30 kmph, the bumps should be close to street lights where it can be seen. A sign showing a speed bump should be placed on the roadside when installing a bump.
- The minimum width of bumps should be 900 mm and height 25 mm. Height should not be more than 100 mm in most cases. In exceptional cases when the traffic needs to be brought to a near halt, 150mm high speed bump may be used.

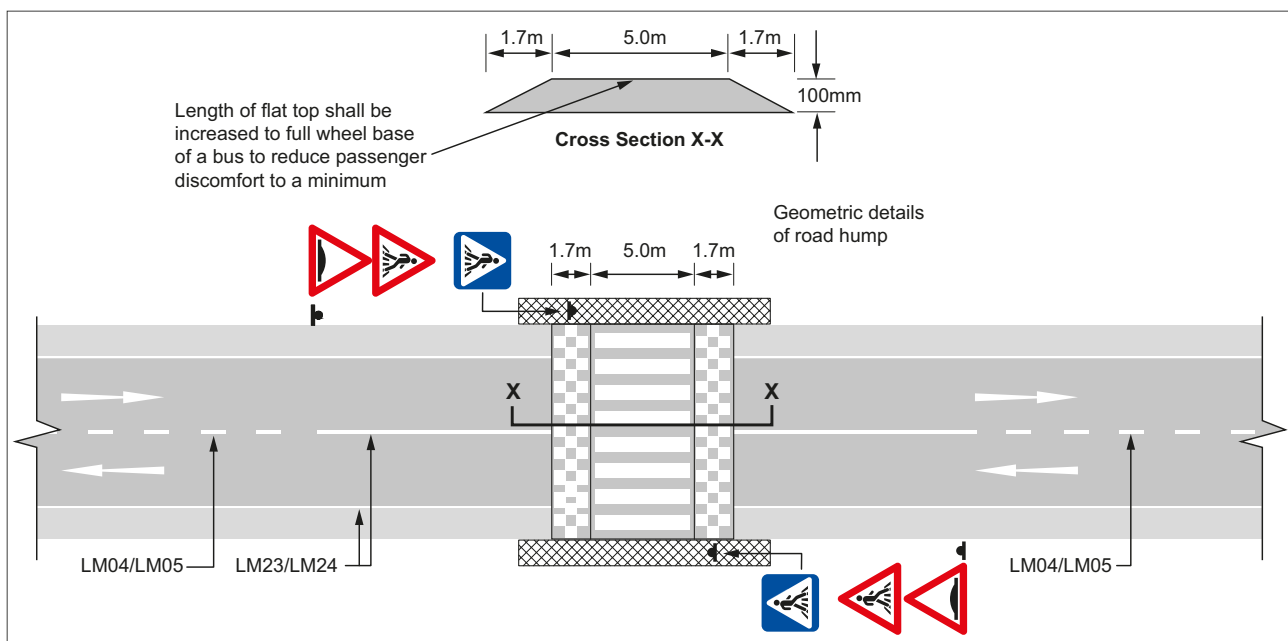
The manufacturer's installation instructions should be strictly complied with. For details of speed bumps, refer to KS 774, specifications for road speed bumps. Traffic should be controlled by traffic controllers during installation, maintenance or removal of bumps.

2.5 Speed Table / Raised Pedestrian Crossing

In road sections where a significant number of pedestrians cross at mid-block location, raised pedestrian crossings or Speed Tables can be provided. Speed Tables give indication to the vehicle drivers about a pedestrian crossing ahead and due to its raised nature, encourages the drivers to reduce speed. The vertical deflection in the Speed Table is to reduce traffic speed and the flat-topped portion is for pedestrians to cross. **Figure 2.4** gives the geometrical details of a Speed Table.

Apart from busy road sections, Speed Tables can also be provided in residential areas as a means of traffic calming and providing safe crossing for pedestrians. The location of Speed Tables should be judiciously selected to provide the facility to a large number of pedestrians.

Figure 2.4 Geometric Details of a Speed Table



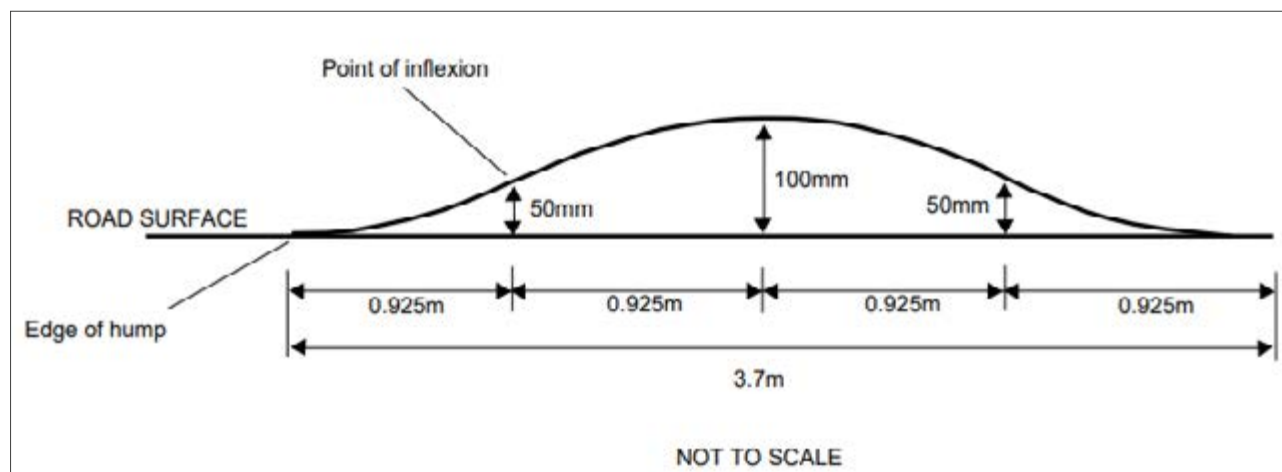
NOTE: For marking details LM04, LM05 etc., refer to RDM 6.1 Road Marking Manual

¹ Source: Traffic Calming – Sinusoidal, 'H' and 'S' humps, TRL Report 377, TRL Limited, UK

2.6 Sinusoidal Speed Hump

Speed humps with a sinusoidal profile are similar to round-top humps but have a flatter initial rise. These provide a more comfortable ride to the cyclists. For motor vehicle drivers, the difference to a circular hump is not significant. The humps can be made of asphalt. Figure 2.5 shows the cross sectional and plan view of sinusoidal humps.

Figure 2.5 Cross section of Sinusoidal Hump



Source: Traffic Calming – Sinusoidal, 'H' and 'S' humps, TRL Report 377, TRL Limited, UK

2.7 Transverse Bar Marking

Transverse Bar Marking is a softer treatment to reduce traffic speed. One set of bar marking comprises of 6 bars, each of 300 mm width, set 600 mm apart on roads with approach speed up to 50 km/h. For roads with higher approach speed, the marking should be placed 1000 mm apart. Each strip is 5 – 10 mm high and should be applied across the full width of the carriageway. The number of sets to be used depends on the approach speed. The distances of placement of the marking from the hazard are given in Table 2.3. The placement of a set of marking is shown in Figure 2.6 and Figure 2.7. Thermoplastic material is to be used for the markings. The colour of bar marking should be yellow on class A and B roads. For all other classes of roads, white colour bar marking is to be used.

Figure 2.6 Traverse Bar Marking

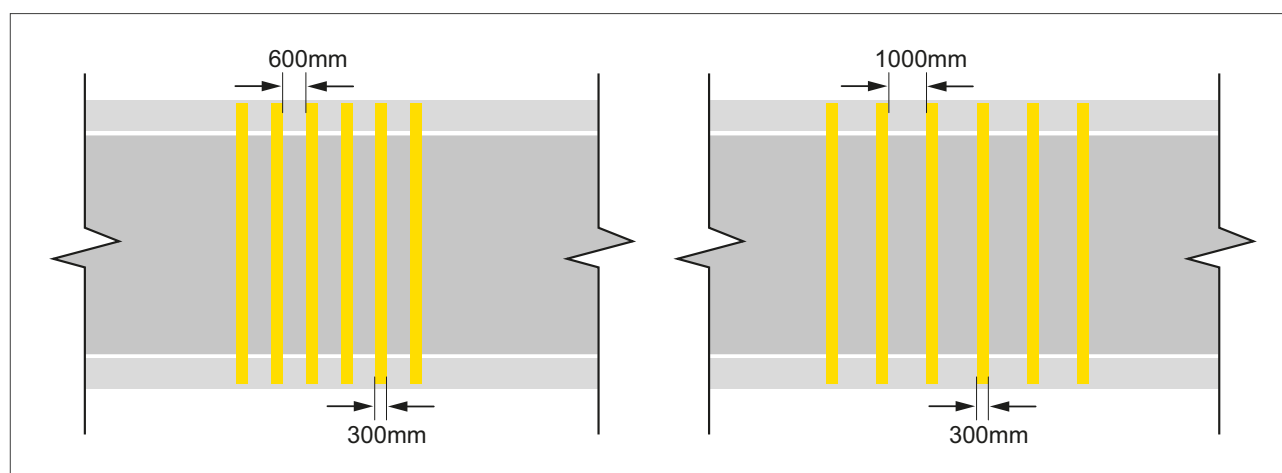


Figure 2.7 Placing of Transverse Bar Marking

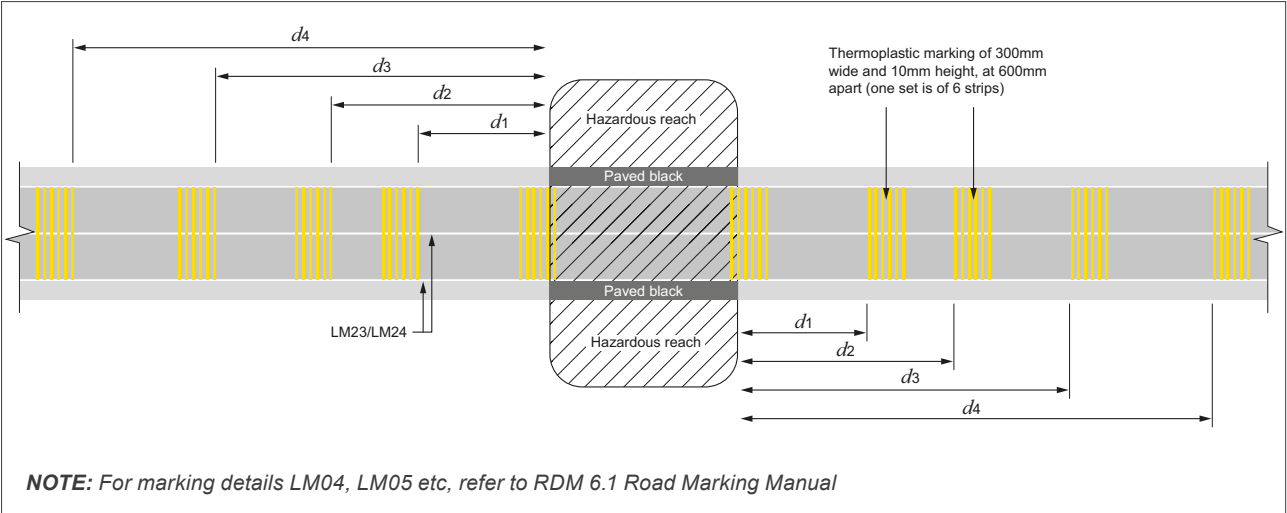


Table 2.3 Number of Bar Marking Sets

Approach Speed (km/h)	Bar Marking (Number of sets)	Distance ($d1$, $d2$, $d3$ & $d4$) from Hazard (m)
Up to 50	1	$d1 = 50$
51 to 65	2	$d1 = 50$, $d2 = 80$
66 to 80	3	$d1 = 50$, $d2 = 80$, $d3 = 120$
81 to 100	4	$d1 = 50$, $d2 = 80$, $d3 = 120$, $d4 = 180$

3 Variable Message Signs (VMS)

3.1 Types of VMS

Variable Message Signs (VMS) are used to inform and direct motorists of situations on the road. The messages are for traffic control and for providing timely information. VMS can be used to give real time route information to the motorists. The messages should be clear and easy to understand. Outdated or incorrect messages should not be displayed.

VMS are designed to assist the driver. This improves traffic flow and operations on the road. Information displayed on a VMS can be for a planned or an unplanned event. The messages are programmed by operations personnel. VMS are usually installed on overhead sign bridges, on post-mounted signs near road shoulders and on overhead cantilever structures. Portable VMS can be mounted on trailers.

The type of information that can be displayed on VMS are:

- Travel time between destinations.
- Information on routes, diversions, congestion, closure.
- Construction warnings.
- Information on special events.
- Maintenance operations.
- Severe weather event announcements.
- Incident notifications.

The objective of VMS display is to assist the driver in taking decisions to take the proper route, avoid congestion, incidents and unfavourable conditions.

VMS can have continuous or discontinuous signs. Continuous signs are similar to fixed signs, the difference being, they can show different messages. Example are: roller blinds, rotating prisms.

Discontinuous signs create various messages on the same sign face using individual elements. Examples are: LED messages, fibre optic signs. VMS used for traffic management are mostly discontinuous LED type.

VMS are a part of Advanced Traffic Management System (ATMS) which is a part of Intelligent Transport System (ITS). ATMS can receive input from various sources and display the processed information on VMS. Input to VMS can be provided manually or by programming.

3.2 Principles

The messages on VMS should meet the following requirements:

1. The message must serve a need.
2. It must command attention.
3. It must be clear and simple.
4. It must give sufficient time for responding to the situation.
5. It must be reliable and credible.
6. It must command respect of road users.

Each VMS message must be for a specific purpose. It should not be used for advertising or public service announcements. Messages posted to warn about road condition or weather condition must

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be removed after the event has passed so that the messages are relevant and credible. Similar type of messages should be used for similar type of event, for continuity and ease of understanding. Examples of messages are given in [Appendix 1](#).

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3.3 Purpose

The purposes of using VMS are: to control, warn and inform.

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3.3.1 Control

VMS messages can be used for speed or lane control, such as lane closure, merging, diverging. Speed control can be achieved by displaying the speed of vehicles.

3.3.2 Warning

The warning messages are for road works ahead or for weather events such as fog, rain and flood.

3.3.3 Information

Information messages can be for incidents such as crashes, congestion, road closure, diversion. These messages must be in large text panels with two or three lines of text. The text can be accompanied by a picture.

3.4 Usage

The situations where VMS messages can be used are: incidents, information, test messages, special events, public announcements.

3.4.1 Incidents

Incidents are instances of some event occurring. These events can be road crashes, diversions, road work, etc. Information on various types of incidents can be displayed on VMS.

Crashes

Information on crashes that block lanes for a period of time can be displayed on VMS. VMS near the crash location can advise motorists about the incident and ask them to change lanes. VMS further away from the location can ask motorists to take alternative routes. Information on minor crashes that occupy travel lanes for only a short time do not need to be displayed.

Diversions

Traffic diversion usually takes place when a road is closed due to severe weather conditions, road works, major incidents, movement of oversized vehicles or movement of very important persons. Messages for such diversions can be displayed on VMS at multiple locations.

Incident Management

Incident management plans are generally prepared by the road authority at regional, corridor and project level. These plans can be activated by the designated officer of the road authority, when required. VMS messages can be used to display messages and routes as per approved plans.

Road Works

VMS messages can be used to warn motorists of ongoing road works or upcoming roadworks. The messages can be for lane closure, lane shift, change of flow from two-way to one-way, work on shoulder, detour, etc. These messages can be used to supplement the road work signs given in the Road Signs Manual.

Weather and Road Condition

Messages can be displayed to warn the motorists of adverse road conditions ahead. These could be: slippery road, flooding or water logging, landslide, mud, rocks, high winds, snow or ice, etc.

3.4.2 Operational Control

VMS can be used to control movement on a lane. Usually, these are used in tunnels and toll plaza. A VMS with a green arrow is placed above the lane that is to be used. A VMS with red cross sign is placed above the lane that is not to be used.

3.4.3 Information

Messages that can be used are: time to reach destination, weather condition, emergency contact numbers, caution for events like flood, general strike, curfew, closure by Police, etc.

3.4.4 Text Messages

These messages can be used during maintenance work. These are general purpose messages, to be used for a short time period.

3.4.5 Special Events

These messages are for planned special events that will affect regular flow of traffic. The messages can be used up to one week prior to the event on the major highways and corridors.

3.4.6 Public Announcements

These messages can be displayed for a limited period so that the main objective of VMS is not affected. Public announcements should not be used during the peak period in the urban areas. Information on public schemes is not appropriate for use. Advertisements should not be displayed on VMS.

Messages related to road safety campaigns can be displayed if media like radio, TV, newspaper, and billboards announce such events. However, such VMS messages should be used sparingly. Display time should be limited to 2 hours in a day, in a staggered manner.

When information display is not required, the VMS should be in a blank mode. When traffic and road conditions are normal, information does not need to be displayed on a VMS.

3.5 Content

VMS provide a versatile means of communication for addressing varying situations. The message must be concise and clear for the motorists to interpret at high speed. Each VMS board must be capable of displaying alphanumeric characters in English and local language. Pictorial signs available for regular traffic signs can be used as VMS.

3.5.1 Timing

The length of time for displaying a message depends on the reading time and exposure time. Reading time is the time taken by the driver to read a message. Exposure time is the time period when the driver is within a legible distance from the sign. Exposure time should be higher than the reading time. Reading and exposure time depends on the approach speed. The message should be designed so that it can be read and understood by the driver at the speed at which they are driving.

The VMS should be legible at 300 m distance on expressways and at 200 m on other roads.

Table 3.1 gives the time it takes to travel 200 m and 300 m at various speeds. It can be used as a guidance for providing for determining the length of a message display.

Table 3.1 Time to Travel 200 m and 300 m

Speed (km/hr)	Time (secs) to Travel 200 m	Time (secs) to Travel 300 m
30	24.0	36.0
40	18.0	27.0
50	14.4	21.6
60	12.0	18.0
70	10.3	15.4
80	9.0	13.5
90	8.0	12.0
100	7.2	10.8
110	6.5	9.8
120	6.0	9.0

For expressways, minimum display time of 20 seconds should be provided for reading and understanding the signs. For other road types minimum 15 seconds should be provided.

The message should be displayed alternatively in English and local language. Simple pictorial messages are easy to read and understand. If the VMS can display 2 lines of message, the first line should be in English, the second line in local language. Each line should have 12 – 15 characters.

When a series of messages are used on a VMS, it is recommended that at least 4 seconds be given to each message. A blinking feature can be used but for just one line of message at a time.

Table 3.2 gives the number of message panels that can be displayed for different approach speeds, for 300 m sight distance. For approach speed less than 70 km/h, maximum of 3 panels can be used.

Table 3.2 Number of Message Panels

Speed (km/hr)	Number of Panels
70	3
90	3
100	2
120	2

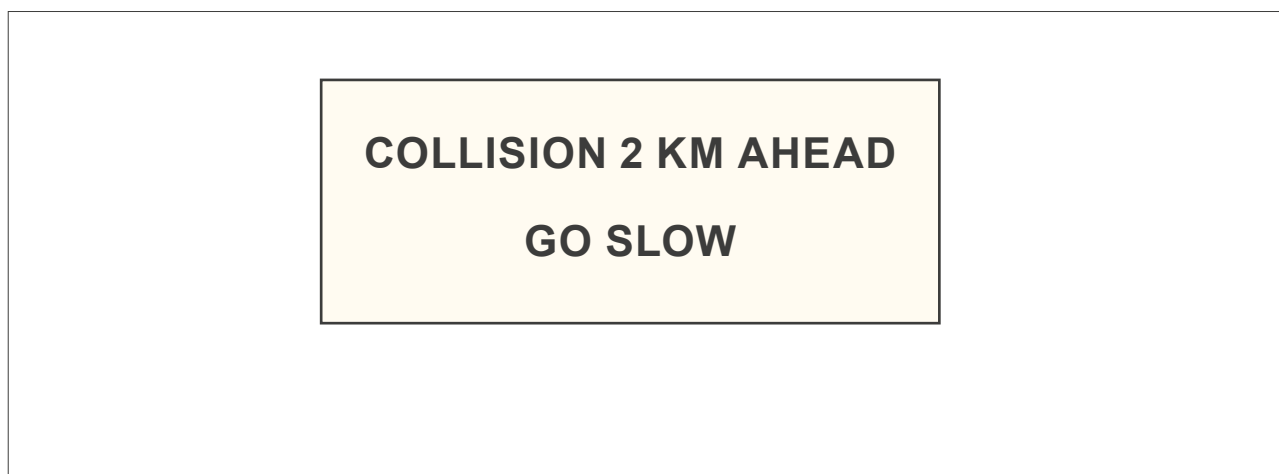
For a single message, steady or blinking message can be used. For blinking message, the message can be on for 2 seconds and off for 1 second.

3.5.2 Height of Lettering

The height of letter should be 400 mm. The manufacturer may choose to use full matrix or line matrix display for displaying vowels.

3.5.3 Message Units

Each message has units of information. Each unit is a separate piece of information that the driver needs to understand and take decision on. A unit is usually one to two words but can be up to four words. An example with two units is given in **Figure 3.1**.

Figure 3.1 Example of Message With Two Units**Table 3.3** Information Breakdown

Units of Info	Information	Driver Question	VMS Message
1	Problem / Descriptor	What happened?	COLLISION HIGH WINDS FLOODING
1	Location	Where?	AT MOI AVE 2 KM AHEAD
1	Lane closed	What is closed?	LFT LN CLSD SINGLE LN ONLY
1	Effect	What is the effect on traffic?	TRAFFIC JAM EXPECT DELAY
1	Audience	Who is the message for?	AIRPORT TRAFFIC
1	Action	What is advised?	USE EXPWY USE ALT ROUTE

3.5.4 Panels and Phases

The driver should be able to read the message twice while driving at the posted speed. If multiple panels are used, it becomes difficult for the driver to read. It is advisable not to use more than two panels.

A phase is one frame of a message. Each phase of message should be complete in one panel and must make sense. Single-phase messages should be used whenever possible.

Two-phase messages can be used only when it is determined that the driver has enough time to read the entire message while driving at the posted speed.

3.5.5 Message Length

Message length is the number of words or characters in a message. An average driver travelling at a high speed can read and understand up to 8-word messages of 4 to 8 characters per word. The number of panels used also limits the understanding of messages. Length of messages should be kept as short as possible.

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Variable Message Signs

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Variable Message Signs

3.5.6 Message Familiarity

Message familiarity helps the driver to understand the message. Unusual messages take more time to understand. Messages in common, simple language are easy to understand. The following types of messages are advised:

1. Write days instead of dates, example WED - THURS instead of 11 MAY - 12 MAY.
2. Write the start and end day and time, example MON 9 AM – WED 9 AM.
3. Write route destination instead of only route number.

3.5.7 Message Anatomy

Messages are divided into information units that when read separately or collectively, convey a complete thought or message to the driver. Each unit answers a question that a driver may have about an incident and provides information to help the driver take a decision. Information should be managed in a logical manner. The order is: the problem, location and effect. Table 3.4 explains the message anatomy.

Table 3.4 Message Anatomy

Phase 1			
Units of Info	Information	Driver Question	VMS Message
1	Problem	What happened?	COLLISION
1	Location	Where?	AT MOI AVE
1	Effect	What is the effect on traffic?	TRAFFIC JAM
Phase 2			
1	Audience	Who is the message for?	AIRPORT TRAFFIC
1	Action	What is advised?	USE EXPWY

3.6 Display

3.6.1 Location of Equipment

The location of the display panel should be such that:

1. It can be seen from all the traffic lanes.
2. The driver can read the message and take appropriate action.

3.6.2 Display Requirements

Displays are of three types:

1. Text only.
2. Graphics only.
3. Combination of both.

The display background must be non-reflective.

3.6.3 Language Requirement

The display is to be in at least two languages, English and local language.

3.7 Urban Area VMS

In urban areas, VMS are provided to display messages on the following:

1. Recurrent congestion
2. Non-recurrent congestion
3. Weather related problems
4. Congestion due to special events
5. Route information
6. Speed restrictions
7. Parking information
8. Other special conditions.

Pole mounted VMS equipment is to be placed away from the shoulder and emergency traffic lanes. The placement of messages should be such that direct sunlight does not fall on the display. For overhead VMS, placement requirement is same as for overhead signs.

3.8 Portable VMS

3.8.1 Placement

The display requirements for portable VMS are similar to that of fixed VMS. VMS should be placed such that the motorists have adequate time to react to the message. The VMS must be located prior to major junction points, so that drivers can take a different route. On expressways, VMS should be placed 2 km ahead of an interchange /exit. It must be repeated every 500 m. VMS must be placed 50 m prior to the point of decision. The VMS should be visible for at least 200 m.

The other requirements for placement are:

1. View of the VMS should not be obstructed.
2. VMS should not obstruct other traffic control devices.
3. For multiple VMS, the spacing should be 200 – 300 m.
4. VMS should not obstruct traffic flow.
5. VMS should preferably be placed behind a crash barrier.
6. The display panel should face the driver.
7. The display panel should be tilted by about 5 – 10° from perpendicular to the road centreline.
8. Blank portable VMS should not be kept facing traffic for extended period.

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3.9 Mechanical Components

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3.9.1 General

The sign portion of the VMS system should consist of sign housing, optical systems, internal wiring, controller system and related equipment, terminal strips for interconnecting wire, etc.

Any electronic or electrical device installed in the housing shall be mounted for easy access to the maintenance personnel. The VMS System shall be modular in design so that it is not necessary for a technician to remove or replace discrete components in the field.

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3.9.2 Weather – Tight Enclosure

All front facing windows and access doors shall be sealed or gasketed to prevent water, dirt and insects from entering the interior. Screened ventilation louvres and drains shall be included to eliminate moisture build up due to condensation.

3.9.3 General Temperature Control

VMS need to operate in the temperature range of -34° C to +65° C. The transfer and absorption of heat due to solar radiation shall be minimised by the design of the housing and front face.

The VMS controller shall monitor the temperature and humidity sensors and perform the proper action when required. Temperature and humidity sensors shall be of solid state. Heaters shall be used to control moisture and to prevent condensation from accumulating on the front face. If the temperature reaches +65° C, VMS should be automatically switched off for safety reasons.

3.9.4 Sign Face

The surface area in front of light emitting pixels shall be protected by a polycarbonate face to prevent entry of water, dust and insects. The surface area not in front of light emitting pixels shall be masked to reduce glare and improve contrast. The pixels shall be designed to eliminate ghost effects from sunshine. Aluminium shades shall be provided to protect the pixels from direct sunlight.

3.9.5 Contrast Shield

An aluminium contrast shield shall be used as a border for the front housing for better legibility. The border should be bolted to the housing or be a part of the housing so that light does not leak. The border should be covered by the same protection as the front so that colour difference is not visible to the motorist.

3.9.6 Housing

The VMS housing shall be in compliance with CD 354, Design of Minor Structures and CD 365, Portal and Cantilever Signs/Signal Gantries, Design Manual for Roads and Bridges, UK.

The structural members of the housing shall use an aluminium alloy. The structural framing members shall be welded. Any other sign access that utilises extrusions as the structural members shall be bolted together using stainless steel and aluminium hardware.

The housing shall be of modular design. The housing shall prevent any light leaks.

3.9.7 Exterior Housing Finish

To minimise heat build-up, all sides of the housing including the back, top and bottom shall have maintenance free natural aluminium finish.

3.9.8 Tilting

In order to properly aim the VMS towards the traffic direction, there should be an option to tilt the entire sign housing and front face, by 0° to 10°. The tilting mechanism should be capable tilting at one-degree increments.

3.9.9 VMS Access

Any access panels shall be limited in size so they may be opened or closed by one person. Panels shall be gasketed and sealed to prevent the elements from entering when panels when they are closed. Panels shall include locks to prevent unauthorised access. Access panels shall be supported in their open position by multiple self-locking retaining devices that support the panel assembly in the open position under wind speed of 64 km/h.

Several accesses are possible and should be defined on the project basis. It is important to define the VMS access in accordance with the design of the supporting structure in order to ensure that the mounting arrangement is adequate and possible.

3.9.10 Walk in Access

The VMS enclosure shall provide access for walking to all the components within the sign. An aisle of 64 cm width with non-skid aluminium flooring shall be provided for walking.

Door: The sign shall have one access door opening outward. A stop mechanism shall be provided to keep the door in the open at 90° position. The open door should be able to withstand wind speed of 64 km/h. The door shall be equipped with a proper latching system. A door handle must be provided outside and inside the housing. A door sensor shall be provided and connected to the VMS controller to monitor opening and closing of the door.

Work area: The interior walkway must be at least 610 mm wide and 1800 mm high with no obstructions. The floor should have drainage facility in a way that insects and dust do not get in through it.

Lighting: One 60 W fluorescent light every 2.40 m of housing shall be provided. The light assembly shall be protected by a cage. A manual timing device for at least 2 hours of lighting shall be provided.

3.9.11 Canopy Doors

For signs greater than 1830 mm high, the access doors shall be hinged at the top and at some other intermediate point, so that when the access doors are in the open position, they shall create a temporary shelter that consists of a roof and partial rear wall.

1. When all of the doors are open, the canopy shall protect the electronic equipment and maintenance personnel from the elements. The gap between two adjacent doors shall be no larger than 19 mm.
2. Opening a canopy door shall allow maintenance personnel immediate access to circuit boards and internal sign parts, without having to remove any items in the sign, or the need to use any tools or to remove any device that could be dropped or lost, such as a locking pin or bolt.
3. The Manufacturer shall furnish any catwalks or framing members necessary to offset the back of the sign from the structure, to reach the sign from the roadside and to provide work space and door clearances. The Manufacturer must provide a solid floor (diamond etched catwalk) in locations over active traffic, to allow maintenance personnel to work over live traffic.

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Variable Message Signs

3.9.12 Mounting Structure

Mounting Structure shall use minimum 5.5 m high hexagonal/octagonal MS pole (minimum 300 mm diameter and 5 mm thickness) with minimum 520 mm x 520 mm x 16 mm base plate (with sufficient stiffeners). It shall be painted with one coat of primer and two coats of PU paint. The structure shall be provided with suitable stiffeners and support angles to survive wind speeds of up to 150 km/h. The minimum vertical clearance under the sign shall be 5.5 m to the road surface.

3.10 Equipment

The VMS sign, controller and any interface cabling between the sign and controller shall be considered as one closed system regardless of where the VMS controller is installed relative to the sign. The protocol and command set between the sign controller and VMS sign shall be entirely independent of and not interfere with any communication protocol or command set used to communicate to the control centre or any remote devices such as a laptop or wireless system.

3.10.1 Wiring

Terminations for the wiring between the VMS sign and the VMS controller cabinet shall be made on screw clamp terminal blocks located at one location within the VMS Sign Housing.

3.10.2 Display

LED Pixel Construction

The VMS shall use high intensity LED to display alphanumeric messages. The number of LEDs to be used for a pixel to achieve the required luminance shall be determined by the manufacturer. Pixel size shall be 15 to 22 mm and pixel pitch shall be 22 to 25 mm. Pixel pitch is the centre-to-centre distance between two adjacent pixels.

Message Construction

The characters of a message shall have a minimum height of 400 mm with an allowance of +/- 20 mm. The aspect ratio of characters shall be 7:5 (height : width). The height of each character shall be a multiple of 7 and it will be related to the pixel pitch. For 14 pixels, at minimum 22.5 mm pixel pitch, the height of the character shall be 315 mm.

Spacing shall be in terms of stroke (S), where $S = 1/7 \times (\text{character height})$. Spacing between two characters shall be 2S and spacing between two lines shall be 4S.

Display Properties

The luminous intensity should be uniform across the display. The supplier shall submit a test certificate from an independent laboratory/Agency to certify that the luminance intensity is at least 12,000 cd/m² under 40,000 Lux. The luminance intensity shall be measured in front of the sign. The brightest pixel shall not be more than 3 times brighter than the least bright pixel. The display shall be designed to automatically adjust the intensity with the ambient brightness based on Light Dependent Resistor (LDR).

Display Change Time

Change from one message to the next shall take place smoothly so that the driver can see the complete message on the sign face.

3.10.3 VMS features

VMS should have following features:

1. Hardware capable of storing 10 sets of messages and associated software capable of sending 10 sets of messages.
2. Emergency message over-riding.
3. Set of messages for a prescribed duration.
4. Temperature monitoring and sending the information to the controller.
5. Serial port for connection to control room or local laptop.
6. Diagnostics features in hardware and software.
7. Modular design for easy replacement.
8. Pictograms of suitable size and hardware for storing 24 such messages.
9. LEDs shall be non-tinted, non-diffused, high intensity, capable of operating in the temperature range of -40° C to +85° C.

3.10.4 VMS Controller

Controller Cabinet

A cabinet shall be provided for all the electronic components and the controller. This cabinet shall be mounted on the floor near the VMS or on a supporting pole.

The cabinet shall be made of aluminium and shall comply with IP55 protection. The cabinet shall include the following:

1. Microprocessor based CPU.
2. The main power supply and energy distribution system.
3. One work lamp to illuminate the work area, when the cabinet door is open.
4. Lightning protection and terminations for the communication and control cables.
5. Termination blocks for the control cables to and from the VMS housing.
6. Permanently mounted document holder.

Electronics

The controller shall be a 485 mm rack mounted microprocessor-based CPU. The controller shall be capable of operating fully, without the need for an external modem or signal booster, within 1 km of the sign.

The controller shall be capable of receiving 12 V DC power supply from a battery, during a power failure and shall be capable of charging the battery.

The Controller shall include a Central Processor Module with a battery backed clock calendar to record power failures and to perform all calculations and logical functions of the VMS Controller.

One VMS Controller shall be capable of operating multiple VMS.

The VMS Controller shall include one integrated digital I/O board with 4 inputs and 4 outputs, with capacity to for add external boards.

1

The VMS Controller CPU shall be at least a 32-bit processor, operating at 400 MHz or higher frequency. It shall have at least 20 MB of SRAM, and be capable of expanding its memory at least 16 MB more, via use of an industry standard USB 'FLASH' memory card.

2

The VMS Controller shall have the ability to control a TFT colour screen with keypad, for local diagnostics and control of the sign. The screen and keyboard shall be at the front of the controller.

3

The VMS Controller shall monitor the sign display, independent of any external commands, and shall cause the signs to display all the appropriate characters.

The VMS Controller shall command the sign to display characters and messages by turning the LEDs on and off. In addition, it shall accumulate data about the status of the sign and shall receive commands from the Central Computer and Portable Maintenance Computer.

3.11 Interfaces

Each VMS Controller shall be uniquely addressable from the Control Centre via the communication system for control and monitoring. The communication profiles that must be supported by the controller are:

PMPP - NULL, PPP = NULL, Ethernet - UDP/IP, RS-232

The VMS Controller shall be capable of communicating using NTCIP or other equivalent international protocols.

3.12 Surge Protection

The VMS electronics should be protected against high voltage surges coming on the field cables, by using appropriate surge arresters. Suitable earthing should be provided.

3.13 Data Storage

Each individual VMS shall be able to store all the required data for displaying information locally. The equipment shall be capable of storing minimum 20 frames that can be triggered on receiving the command.

3.14 Data Communication

Data Communications may be through any of the links such as dedicated line, leased line from the local service provider, GSM/CDMA DATA channel, GSM/CDMA SMS channel.

Data communication shall be provided with sufficient security check to avoid unauthorised access.

3.15 Connectivity

Each VMS unit shall be provided with a unique identification number and shall communicate with the designated Central Control Computer only. Sufficient security checks shall be implemented before accepting message for storage or display. VMS unit shall respond to command, generated by Central Control Computer, for communication link diagnostics.

The VMS unit shall be able to accept messages created on Central Control Centre, using Graphical User Interface application. Messages shall be accepted either for storage or display. It shall display various message attributes such as message blink, blank, entry style (left, top, right entry).

The VMS unit shall be able to send its health status to the Central Control Centre on query basis. This query shall be generated by Central Control Computer and transmitted to VMS unit. The health status information shall include individual display matrix LED status, Controller status, etc. All this information shall be time and date stamped for proper MIS report generation.

3.16 Power Requirement

The VMS shall be powered by 230 V AC, 50 Hz single phase power supply. The equipment components shall have adequate surge and lightning protection.

Inverter of adequate power capacity with 12-hour back up shall be provided, to account for power failure.

3.16.1 Design Life

The minimum design life for the equipment before replacement shall be 10 years.

3.17 Central Control Computer

The central control computer shall be capable of setting an individual VMS, or group of VMS, to display either one of the pre-set messages, or symbols, or a text message. In addition, the central control computer system shall be capable of being programmed to display an individual message on a VMS, or a group of VMS, at a pre-set date and time. A sequence of a minimum of 10 messages / symbols per VMS shall be possible.

The central control computer shall also store, in a suitable database, information about the messages displayed on each VMS. The minimum information to be stored shall be:

1. The identification number of the VMS.
2. The message/symbol content, or the standard message/symbol number.
3. The start date and time at which the message/symbol was displayed.
4. The end date and time at which the message/symbol was displayed.

The central control computer shall test each individual VMS on a regular (pre-set) basis. This test should be exhaustive and determine if the system is fully operational.

3.18 General Requirements

1. Housing: Powder coated housing with ingress protection IP55 or other equivalent international standards for protection against dust, rain and wind.
2. The structure on which the VMS will be mounted shall be sturdy and aesthetically designed and capable of bearing wind loads up to 200 km/h.
3. Protection against EMI: The electronic circuitry and wiring inside the VMS shall be protected against any kind of EMI.

3.19 Specific Requirements of Portable VMS

3.19.1 Storage

Each individual VMS shall be able to store all the required data for displaying information locally. The equipment shall be capable of storing minimum 10 frames that can be triggered on receiving the command.

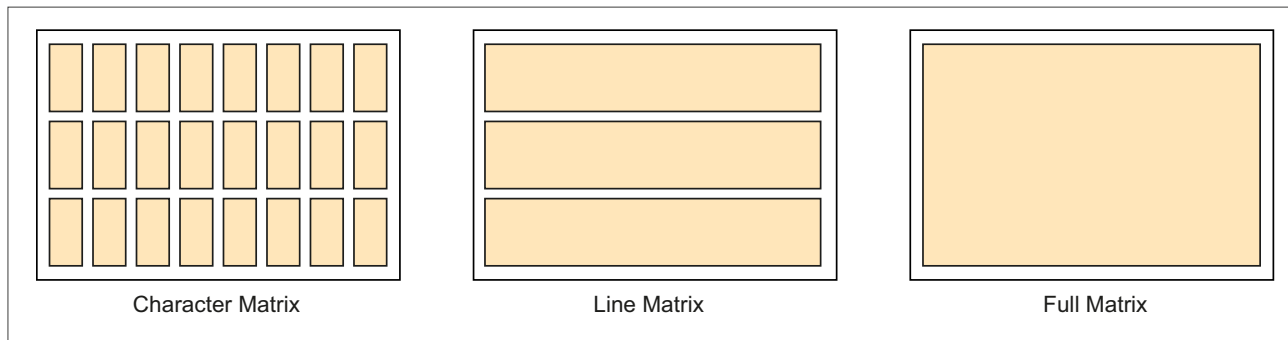
Equipment shall be portable and capable of being installed on a vehicle.

3.20 Design of VMS

3.20.1 Matrix Displays

There shall be three typical types of matrix display: character, line, and full. Figure 3.4 the three matrix types. Line and full matrix are recommended.

Figure 3.2 VMS Matrix Displays



Messages shall be displayed by using single or multiple-phases. A phase is defined as the limit of the display area available for text, bitmaps, or animation. Messages that require more information to be displayed can be shown in multiple phases.

3.20.2 Design Process

The design process presented below shows the steps needed for a successful deployment of VMS.

1. Collect preliminary data regarding the purpose of VMS, type of information to be displayed and the alternative routes.
2. Determine VMS type based on purpose and type of information to be displayed and technology to be used.
3. Determine placement location of VMS.
4. Collect site-specific data required for the proposed location.
5. Select the VMS site required for design.
6. Determine cabinet placement location.
7. Prepare underground connection infrastructure.
8. Determine communication medium for location.
9. Deploy VMS.

References

1. **AASHTO**, Standard Specifications for Structural Supports for Highway Signs, Luminaires and Traffic Signals, 6th edition.
2. **KS 774**, Specifications for road speed bump.

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References

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References

Appendices

Appendix 1: Typical VMS Messages

Closure	
ACCIDENT AHEAD ROAD CLOSED	ROAD CLOSED 2 KM AHEAD
LEFT LANE CLOSED AHEAD	RIGHT LANE CLOSED AHEAD
EXIT CLOSED	REST AREA CLOSED
TUNNEL CLOSED AHEAD	RAMP CLOSED
Construction	
BRIDGE WORK AHEAD	SHOULDER WORK AHEAD
NIGHT WORK AHEAD	TRUCKS CROSSING
CONSTRUCTION AHEAD EXPECT DELAYS	ROAD PAVING WORK AHEAD
CONSTRUCTION NEXT 3 KM	SHOULDER WORK AHEAD
Direction	
KEEP LEFT	ACCIDENT AHEAD ALL TRAFFIC MUST EXIT
KEEP RIGHT	ACCIDENT AHEAD MERGE LEFT
LANE CLOSURE AHEAD EXPECT DELAYS	MERGE LEFT
DETOUR	FORM ONE LANE LEFT
EXPECT DELAYS	HEAVY TRAFFIC AHEAD
Weather	
DENSE FOG AHEAD	FOGGY CONDITIONS AHEAD
HIGH WIND ADVISORY	WATER ON ROAD
ROAD FLOODED AHEAD	REDUCED VISIBILITY AHEAD

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Appendices